Distilled retrieval student: Methodology and Evidence

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Status. Post-run methodology reconstruction grounded in archived source and reports. This document is not an executed historical notebook.

Research question

Compress the full retrieval model's ranking signal into a smaller student. The GPU report names full as teacher and lists about 67.8 million trainable student parameters.

Architecture and mechanism

The full model supplies teacher targets. A lower-capacity student learns to rank the same candidate pools; its outputs are exported with the self-match rule enforced.

Measured result

Scored submission	1039888
Codabench micro-F1@7	0.4034
Archived local metric	0.3969 corrected local-dev micro-F1@7

Corrected local dev micro-F1@7 was 0.3969 and Codabench was 0.4034. The hidden result is close to the full model but does not exceed it.

Limitations and next check

Teacher-target quality and distillation temperature/loss affect this comparison; the reported score alone cannot isolate them. Local, pilot, and hidden metrics are deliberately kept distinct. A single scored run does not establish a causal architecture effect or statistical significance.

Source and reproducibility

Original source modules: phase7_run.py, phase7_export.py. SHA-256 hashes, the original report, and the prediction ZIP receipt are in the companion provenance file. The explanatory notebook was written after the historical run; original execution output is never implied. Reproduction requires authorized course data, the Phase 1 parent weights/tokenizer, and the dependencies in the archived source.

Documentation disclosure

AI assistance was used to organize the archived evidence and draft this explanatory document. All numerical claims above come from saved run records or the provided finished-submission screenshot; missing evidence is labelled as such.